
Questions from February 21, 2025

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Question 1

I have written " Uniform converges of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 1.7 (this lemma states for $F_m = \text{supp } \psi_x^{(m)}$ then $F_0 \supset F_1 \supset F_2 \supset \dots$) we can choose m large enough so for all $x, y \in F_m$ that $|f(x) - f(y)| < \epsilon$, then

$$\begin{aligned} \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - \frac{\int_K f(x) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &= \left| \frac{\int_K (f(y) - f(x)) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &\leq \frac{\int_{F_m} |f(y) - f(x)| d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &\leq \frac{\epsilon \int_{F_m} d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &= \epsilon \end{aligned}$$

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We talked about F_m being compact, but i find that F_m is open and therefore not compact. This is since we remove the end points from F_m namely the points in V_m .

But do we even need F_m to be compact, can't we just use continuity of f to prove it as i have done above?

Question 2

I think i can solve the convergence of $\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y)$, but i think I am getting confused by the notation.

We have gotten

$$\begin{aligned} \mathcal{E}_m(u, v) &= - \bigotimes_{x \in V_m \setminus V_0} v(x) \left(\int_K \psi_x^{(m)}(y) d\mu(y) \right) r^{-m} \left(\int_K \psi_x^{(m)}(y) d\mu(y) \right)^{-1} \Delta_m u(x) \\ &= - \int_K \left(\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) \right) d\mu(y). \end{aligned}$$

When showing that

$$\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) \rightarrow v(x) f(x) \quad \text{as } m \rightarrow \infty$$

But here i would still say that

$$\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) = v(y) f_m(y)$$

and from this point it is just taking the limit.

But we talked about that the proof of this convergence would be similar to the one for $\int_K f \psi_x^{(m)} d\mu$, so i think i am doing something wrong.

Again i am sure this is because of my misunderstanding of $\psi_x^{(m)}$ and the notation of the book, but i cannot quite figure out how i am suppose to understand this.